30-DAY GIT LEARNING SERIES

GIT QUEST

LEVEL 7

WEEK 1 COMPLETE!

Week 1 Recap

Quiz and Cheat Sheet | Days 1 to 7

QUEST PROGRESS: DAY 7 OF 30

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Your Week 1 Journey

Seven days, seven new skills. Here is the whole map.

DAY 1

Git vs GitHub

Git saves your work. GitHub shares it.

```
git --version
git config --global
```

DAY 2

git init + first commit

Open a save slot, make a save point.

```
git init -b main
git status
git commit -m
```

DAY 3

The 3 areas

Desk, packing table, cupboard.

```
git add <file>
git commit
git restore --staged
```

DAY 4

Add + good messages

Label every box clearly.

```
git add .
git commit -m "msg"
-m "summary" -m "body"
```

DAY 5

git log

Your time machine.

```
git log --oneline
git log --stat
git show <hash>
```

DAY 6

git diff

Red pen, green pen.

```
git diff
git diff --staged
git diff HEAD~1 HEAD
```

DAY 7

Week 1 recap

Quiz and cheat sheet: you are here!

```
git status
(everything!)
```

WEEK 1 COMPLETE!

7 days, 7 skills

The Whole Workflow, One Picture

Everything from Days 2 to 6 connects like this.

1 WORKING DIRECTORY

Your desk

Where you create and edit files.

`git add`

2 STAGING AREA

The packing table

Files picked for the next save.

`git commit`

3 REPOSITORY

The cupboard (.git)

Sealed save points, kept forever.

`git restore --staged (unpack)`

LOOK-AROUND COMMANDS

`git status`	`git diff`	`git diff --staged`	`git log --oneline`
Which area is each file in?	What did I change but not pack?	What am I about to commit?	What is my history?

THE DAILY LOOP

1. `git status`
2. `git add`
3. `git diff --staged`
4. `git commit`
5. `git log --oneline`

Check, pack, review, seal, read. Repeat all week, every week!

Boss Quiz and Final Quest

Ten questions, then build a repo from scratch.

WEEK 1 BOSS QUIZ: 10 QUESTIONS

1. What is the difference between Git and GitHub?
2. Which command turns a folder into a repository?
3. Name the 3 areas of Git.
4. Which command moves a file from the desk to the packing table?
5. What makes a good commit message?
6. What does HEAD mean?
7. Which command shows one line per commit?
8. In git diff, what does a line starting with + mean?
9. Which command shows what you are about to commit?
10. How do you leave the log screen when it opens?

FINAL QUEST: BUILD IT FROM SCRATCH

- [ ] 1. Make a repo

```
git init -b main week1-final && cd week1-final
```

- [ ] 2. Put two files on your desk

```
echo "One" > a.txt && echo "Two" > b.txt
```

- [ ] 3. Check where they are

```
git status
```

- [ ] 4. Pack ONE file

```
git add a.txt
```

- [ ] 5. Last check before saving

```
git diff --staged
```

- [ ] 6. Seal it with a clear label

```
git commit -m "Add file a"
```

- [ ] 7. Repeat for the second file

```
git add b.txt && git commit -m "Add file b"
```

- [ ] 8. Read your history

```
git log --oneline
```

The Week 1 Cheat Sheet

Screenshot this page. Answer key on the right.

SETUP

`git --version`	Check Git is installed
`git config --global user.name "Name"`	Tell Git who you are
`git init -b main`	Create a repository here

THE DAILY LOOP

`git status`	Which area is each file in?
`git add <file>  |  git add .`	Desk to packing table
`git commit -m "message"`	Packing table to cupboard
`git commit -m "s" -m "body"`	Add a longer note

LOOK AROUND

`git log --oneline`	History, one line each
`git log --stat`	History plus files touched
`git show <hash>`	Open one commit
`git diff`	Desk vs packing table
`git diff --staged`	Packing table vs cupboard
`git diff HEAD~1 HEAD`	Compare two commits

UNDO (SO FAR)

`git restore --staged <file>`	Take a file off the table

ANSWER KEY

1. Git saves work on your computer; GitHub stores and shares it online.
2. git init
3. Working directory (desk), staging area (table), repository (cupboard).
4. git add
5. Short, starts with a verb, no full stop, one logical change.
6. The commit you are on: the 'you are here' pin.
7. git log --oneline
8. The line was added.
9. git diff --staged
10. Press q.

`Your Day 7 commit: days-1-to-7-summary.md  ->  docs(day07): add week 1 recap summary`

LEVEL 8: .gitignore | Week 2 starts tomorrow

Undo and Branch: first stop, telling Git which files to ignore.

LOCKED